Linear Regression

https://pub.towardsai.net/calculating-simple-linear-regression-and-linear-best-fit-an-in-depth-tutorial-with-math-and-python-804a0cb23660

Introduction

- Simplest type of model in machine learning
- Statistical approach - Allows us to study and summarize the relationship between two continuous quantitative variables.
  - Applied in machine learning models, mathematics, statistical modeling, forecasting epidemics, and other quantitative fields.
- Two variables
  - Independent variable (X)
  - Dependent variable (Y)
  - Goal is to find out the **relationship** between independent and dependent variables
- Three types of relationships
  - Deterministic relationship
  - Random relationship
  - Statistical relationship

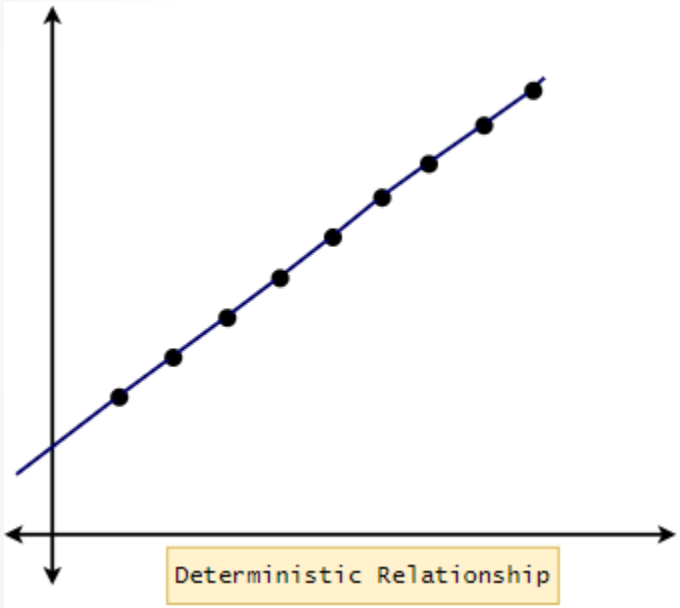
Relationship types

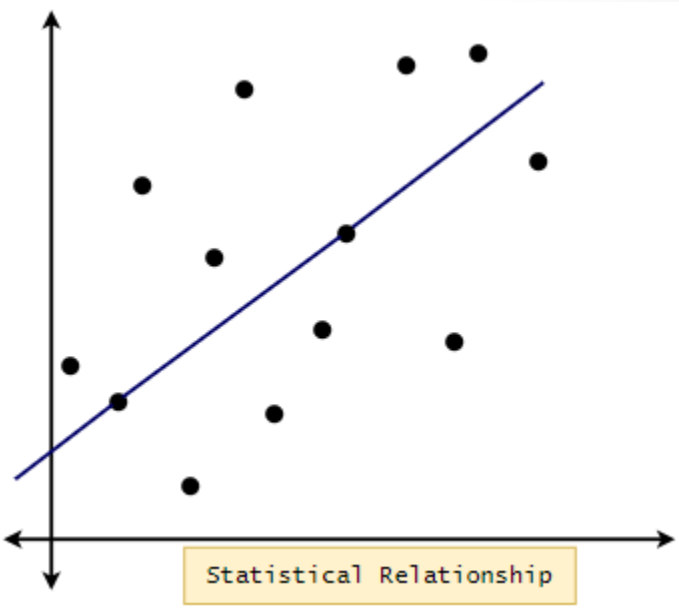
- **Deterministic relationship** - we can predict the output variable using its function
  - Diameter = 2*pi*radius
  - Fahrenheit = 1.8*celsius+32

- **Random relationship -** there are no relationships between the variables

- **Statistical relationship -** relationship that is not so perfect, mixture of deterministic and random relationships
  - Number of chocolates vs. cost
  - Income vs. expenditure

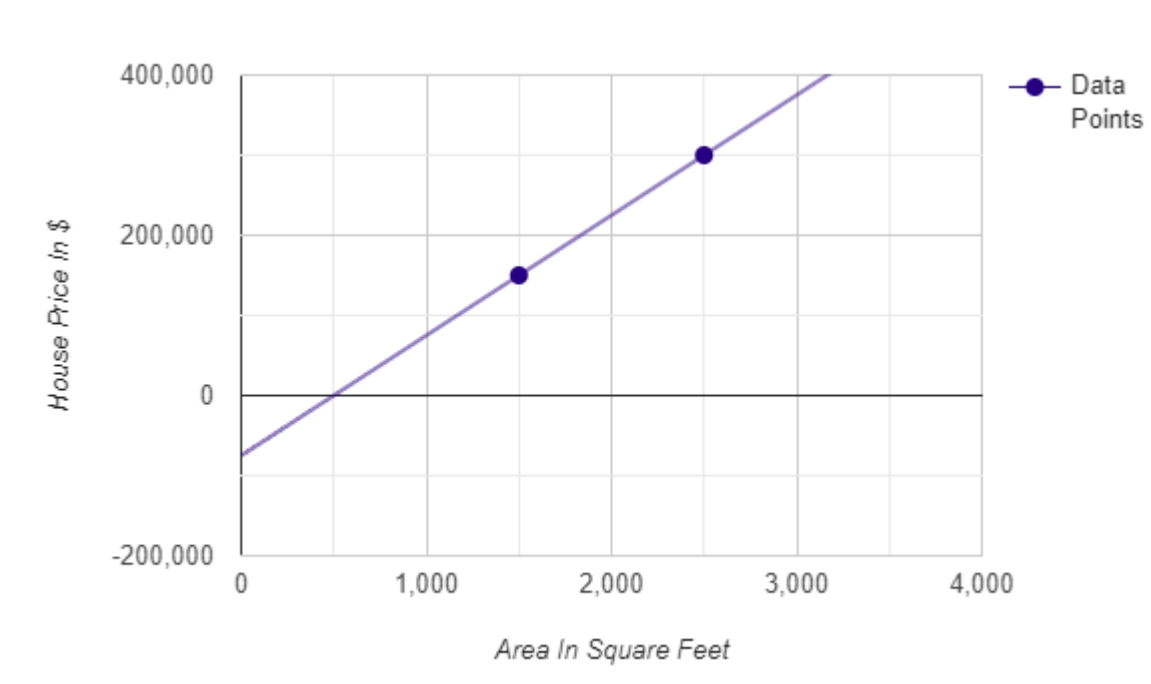
Cont.





Example

Area	Price
1500	150000
1800	???
2500	300000



Area vs. house price on a scatter plot.

Basics of Coordinate Geometry

1. We always look from left to right in the coordinate plane to name the points.
2. After looking from left-to-right, the first point we get must be named (x1,y1), and the second point will be (x2,y2).
3. Horizontal lines have a slope of 0.
4. Vertical lines have an "infinite" slope.
5. If the second point's Y-coordinate is greater than the Y-coordinate of the first point, then the line has a positive(+) slope. Otherwise, the line has a negative slope.
6. Points at the same vertical distance from X-axis have the same Y-coordinate.
7. Points at the same vertical distance from Y-axis have the same X-coordinate.

Cont.

We all know that the equation of the line:

$$Y = mX + b$$

Where,

$m = slope\ of\ the\ line$

$b = Y - intercept\ of\ the\ line$

$Y = Y\ coordinate\ of\ the\ point\ from\ which\ the\ line\ passes\ through$

$X = X\ coordinate\ of\ the\ point\ from\ which\ the\ line\ passes\ through$

$$m = \frac{Y_2 - Y_1}{X_2 - X_1}$$

Where,

$m \quad = slope\ of\ the\ line$

$(X_1, Y_1) = X\ and\ Y\ coordinates\ of\ the\ first\ data\ point$

$(X_2, Y_2) = X\ and\ Y\ coordinates\ of\ the\ second\ data\ point$

Cont.

From the rules mentioned above, we can infer that in our graph:

(X1 , Y1) = (1500 , 150000)

(X2 , Y2) = (2500 , 300000)

Next, we can easily find the slope of the two points.

$$m = \frac{300000 - 150000}{2500 - 1500}$$

$$m = \frac{150000}{1000}$$

$$m = 150$$

Taking our example into consideration, in our equation, Y represents the house's price, and X represents the area of the house.

Cont.

- Now since we have all the other values, we can calculate the value of intercept b.

$$Y = mX + b$$

$$b = Y - mX$$

$$(X, Y) = (1500,150000)$$

$$b = 150000 - 150 * 1500$$

$$b = -75000$$

$$Y = mX + b$$

$$b = Y - mX$$

$$(X, Y) = (2500,300000)$$

$$b = 300000 - 150 * 2500$$

$$b = -75000$$

Cont.

- Next, since we have all our parameters, we can write the equation of line as:

$$Y = mX + b$$

$$Y = 150X - 75000$$

- To find the price of Squidward's house, we need to plug-in X=1800 in the above equation.

$$Y = mX + b$$

$$Y = 150X - 75000$$

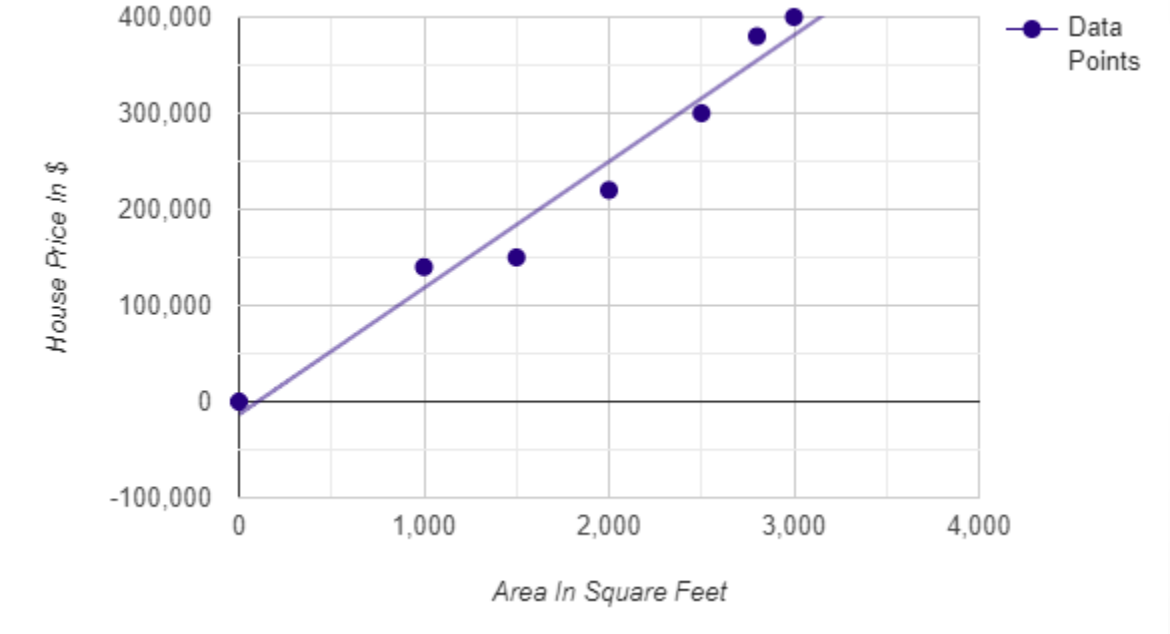
$$Y = 150 * 1800 - 75000$$

$$Y = 195000$$

Things to Notice

- We live in real world
  - Only two data points are insufficient information
  - We need a large dataset to generalize or improve our system's prediction performance
- But in a real world environment, with million of data points
  - Can we find one "*single*" *straight line that* "perfectly" *fits the whole data?*
  - *That's where the concept of best fit/ regression line comes in.*

Cont.

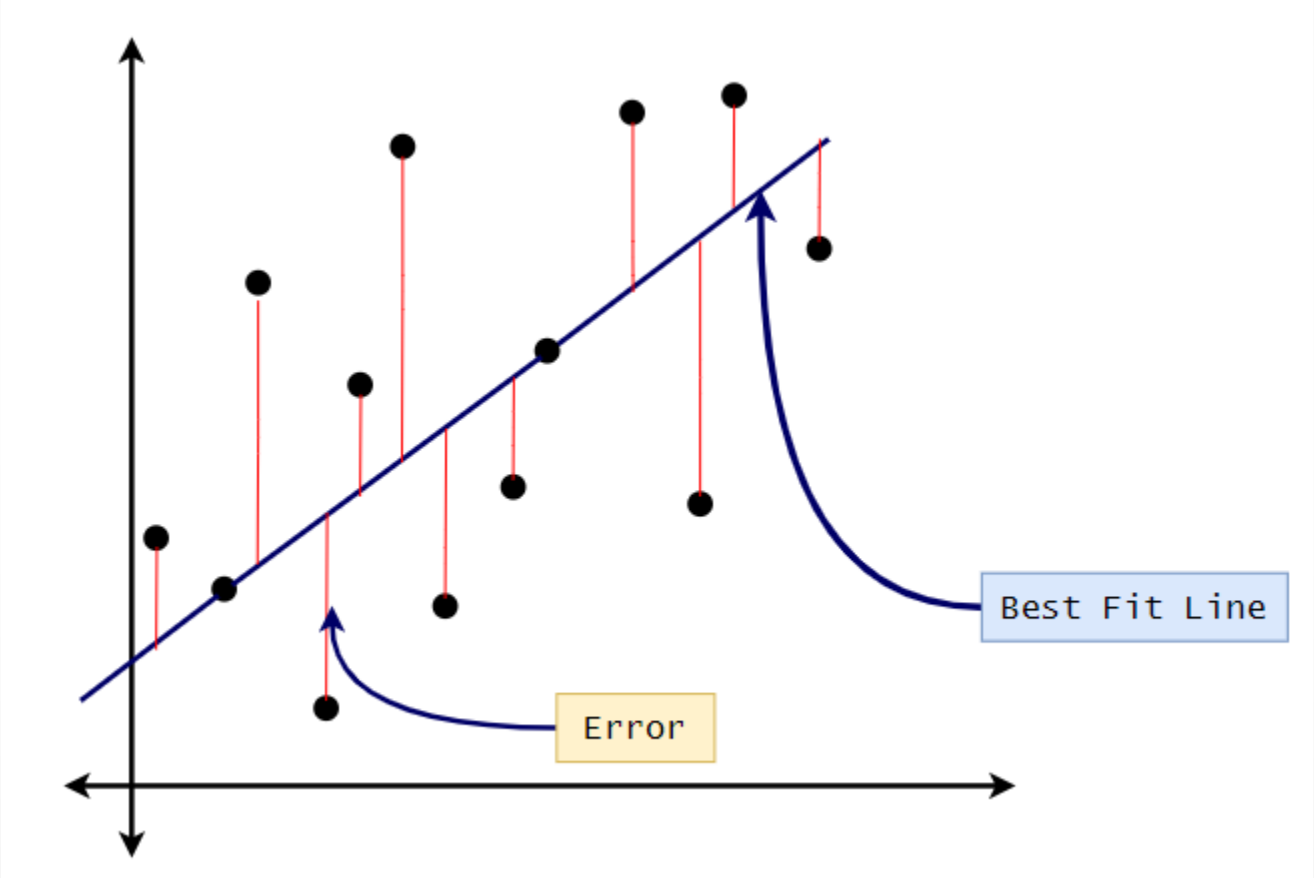


- Central concept remains the same
- Find the equation of the line and plug-in X's value (independent variable) to find Y's value (dependent variable)

Calculating the Linear Best Fit

- We cannot plot a single straight line that passes through all the points
- Let's minimize the error
  - Find a bar and then find the prediction error
- We have the actual value here, we can easily find the error in prediction.
- Goal will be to find the line that has the minimal error.
  - That line is called the **linear best fit**.

Example



Sum of Squared error

- Actual selling price of a house with an area of 1800 square feet is $220,000
- Predict the house price (Y = 150X-75000 = 195,000)
- Here, there is a prediction error
- To represent, we can se the Sum of Squared error calculation technique to find the error in prediction for each of the data points.
- We randomly choose the parameters of our line and then calculate the error.
- Afterward, we will adjust the parameter again and then calculate the error.
- Repeat this until we get the minimum possible error
- This process is a part of the gradient descent algorithm

Error

Positive error:

- Actual selling price: $ 220,000
- Predicted selling price: $ 195,000
- Error in prediction: $ 220,000–$ 195,000 = $25,000

Negative error:

- Actual selling price: $ 160,000
- Predicted selling price: $ 195,000
- Error in prediction: $160,000 — $195,000 = -$35,000

Cont.

- It is also possible to get a negative error. To account for negative errors, we square the error.

$$Errors = \sum_{i=1}^{i=N} (Y_{i\,actual} - Y_{i\,predicted})$$

- To account for the negative values, we will square the errors.

$$Errors = \sum_{i=1}^{i=N} \left(Y_{i\,actual} - Y_{i\,predicted}\right)^2$$

Guidelines for regression line

1. Use regression lines when there is a significant correlation to predict values.
2. Stay within the range of the data, and make sure not to extrapolate. For example, if the data is from 10 to 60, do not try to predict a value for 500.
3. Do not make predictions for a population that base on another population's regression line.

Classical Example Problems

1. Height and weight.
2. Vital lung capacity and pack-years of smoking.
3. The driving speed and gas mileage.

Best fit line issues

- **Issue in the used approach:** Although can find the best fit line for a specific dataset but for a larger dataset, this task can be cumbersome
- **Proposed solution:** use a formula that will give us the required parameter values.
- **Roadmap:** Derivation of the formula followed by its implementation

Derivation of Simple Linear Regression Formula:

1.We have a total of n data points (X, Y), ranging from i=1 to i=n.

$$(X_i, Y_i)$$

2. We define the linear best fit as:

$$\hat{Y}_i = a + BX_i$$

3. We can write the error function as following:

$$S = \sum_{i=1}^{i=n} \left(Y_i - \hat{Y}_i\right)^2$$

4. We can substitute the value of equation 2 in equation 3:

$$S = \sum_{i=1}^{i=n} (Y_i - a - BX_i)^2$$

Minimize Error (S)

- To minimize S (error function), lets find where the first derivative of S is equal to 0 concerning a and b.
- **Finding Intercept (a):**
  1. Finding the partial derivative of S concerning a:

$$\frac{\partial S}{\partial a} = \frac{\partial}{\partial a}\left[\sum_{i=1}^{i=n}(Y_i - a - BX_i)^2\right]$$

  2. Simplifying the calculations:

$$Y_i - a - BX_i = u$$

$$S = \sum_{i=1}^{i=n} u^2$$

  3. Using chain rule of partial derivations:

$$\frac{\partial S}{\partial a} = \frac{\partial S}{\partial u} * \frac{\partial u}{\partial a}$$

  4. Finding partial derivatives:

$$\frac{\partial S}{\partial u} = \frac{\partial}{\partial u}\left[\sum_{i=1}^{i=n} u^2\right] = 2 * \sum_{i=1}^{i=n} u$$

$$\frac{\partial u}{\partial a} = \frac{\partial}{\partial u}(Y_i - a - BX_i) = 0 - 1 - 0 = -1$$

Cont.

5. Putting it together:

$$\frac{\partial S}{\partial a} = -2 * \sum_{i=1}^{i=n} u = -2 * \sum_{i=1}^{i=n} (Y_i - a - BX_i)$$

6. To find the extreme values, we take the derivative=0:

$$-2 * \sum_{i=1}^{i=n} (Y_i - a - BX_i) = 0$$

7. Simplifying:

$$\sum_{i=1}^{i=n} (Y_i - a - BX_i) = 0$$

8. Further simplifying:

$$\sum_{i=1}^{i=n} a = (a + a + a \ldots + a)_{n\,times} = n * a$$

Cont.

9. Finding the summation of a:

$$\sum_{i=1}^{i=n} a = (a + a + a \ldots + a)_{n\,times} = n * a$$

10. Substituting the values in the main equation:

$$\sum_{i=1}^{i=n} Y_i - na - B\sum_{i=1}^{i=n} X_i = 0$$

11. Simplifying the equation:

$$a = \frac{\sum_{i=1}^{i=n} Y_i - B\sum_{i=1}^{i=n} X_i}{n}$$

12. Further simplifying the equation:

$$\frac{\sum_{i=0}^{i=n} Y_i}{n} = \bar{Y}$$

$$\frac{\sum_{i=0}^{i=n} X_i}{n} = \bar{X}$$

13. Simplifying the equation for the value ot a:

$$a = \bar{Y} - B\bar{X}$$

Finding B (Slope)

1. Finding the partial derivative of S concerning B:

$$\frac{\partial S}{\partial B} = \frac{\partial}{\partial B}\left[\sum_{i=1}^{i=n}(Y_i - a - BX_i)^2\right]$$

2. Simplifying the calculations:

$$Y_i - a - BX_i = u$$

$$S = \sum_{i=1}^{i=n} u^2$$

3. Using chain rule of partial derivations:

$$\frac{\partial S}{\partial B} = \frac{\partial S}{\partial u} * \frac{\partial u}{\partial B}$$

Cont.

4. Finding partial derivatives:

$$\frac{\partial S}{\partial u} = \frac{\partial}{\partial u}\left[\sum_{i=1}^{i=n} u^2\right] = 2 * \sum_{i=1}^{i=n} u$$

$$\frac{\partial u}{\partial B} = \frac{\partial}{\partial B}(Y_i - a - BX_i) = 0 - 0 - X_i = -X_i$$

5. Putting it together:

$$\frac{\partial S}{\partial B} = -2\sum_{i=1}^{i=n} X_i * u = -2\sum_{i=1}^{i=n} X_i * (Y_i - a - BX_i)$$

6. Distributing Xi:

$$\frac{\partial S}{\partial B} = -2\sum_{i=1}^{i=n}(X_iY_i - X_ia - BX_iX_i)$$

Cont.

7. To find the extreme values, we take the derivative=0:

$$-2\sum_{i=1}^{i=n}(X_iY_i - X_ia - BX_i^2) = 0$$

8. Simplifying:

$$\sum_{i=1}^{i=n}(X_iY_i - X_ia - BX_i^2) = 0$$

9. Substituting the value of a in our equation:

$$a = \bar{Y} - B\bar{X}$$

$$\sum_{i=1}^{i=n}(X_iY_i - X_i(\bar{Y} - B\bar{X}) - BX_i^2) = 0$$

Cont.

10. Further simplifying:

$$\sum_{i=1}^{i=n}\left(X_iY_i - \bar{Y}X_i + BX_i\bar{X} - BX_i^2\right) = 0$$

11. Splitting up the sum:

$$\sum_{i=1}^{i=n}(X_iY_i - \bar{Y}X_i) + \sum_{i=1}^{i=n}(BX_i\bar{X} - BX_i^2) = 0$$

12. Simplifying:

$$\sum_{i=1}^{i=n}(X_iY_i - \bar{Y}X_i) - B\sum_{i=1}^{i=n}(X_i^2 - X_i\bar{X}) = 0$$

Cont.

13. Finding B from the above equation:

$$B = \frac{\sum_{i=1}^{i=n}(X_i Y_i - \bar{Y} X_i)}{\sum_{i=1}^{i=n}(X_i^2 - X_i \bar{X})}$$

14. Further simplifying the equation:

$$B = \frac{\sum(XY - \frac{\sum Y}{n} X)}{\sum(X^2 - \frac{\sum X}{n} X)}$$

$$B = \frac{n\sum XY - \sum Y \sum X}{n\sum X^2 - \sum X \sum X}$$

$$B = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$$

Generalized form for Intercept (a)

- Get the value of a (intercept)

$$a = \bar{Y} - B\bar{X}$$

$$a = \bar{Y} - \left(\frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}\right)\bar{X}$$

$$a = \frac{\sum Y}{n} - \left(\frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}\right)\frac{\sum X}{n}$$

Cont.

- Simplify the formula:

$$a = \frac{1}{n} * \frac{[\sum Y * (n\sum X^2 - (\sum X)^2) - (n\sum XY - \sum X \sum Y)\sum X]}{n\sum X^2 - (\sum X)^2}$$

$$a = \frac{1}{n} * \frac{n\sum X^2 \sum Y - (\sum X)^2 \sum Y - n\sum XY \sum X + \sum X \sum Y \sum X}{n\sum X^2 - (\sum X)^2}$$

$$a = \frac{1}{n} * \frac{n\sum X^2 \sum Y - (\sum X)^2 \sum Y - n\sum XY \sum X + (\sum X)^2 \sum Y}{n\sum X^2 - (\sum X)^2}$$

$$a = \frac{1}{n} * \frac{n(\sum X^2 \sum Y - \sum XY \sum X)}{n\sum X^2 - (\sum X)^2}$$

- $$a = \frac{\sum X^2 * \sum Y - \sum X * \sum XY}{n\sum X^2 - (\sum X)^2}$$

Summary

Simple Linear Regression Formulas

$$Best\ Fit\ Line: \hat{Y}_i = a + BX_i$$

$$a = \bar{Y} - B\bar{X}$$

$$B = \frac{\sum(X_i Y_i - \bar{Y} X_i)}{\sum(X_i^2 - X_i \bar{X})}$$

$$a = \frac{\sum X^2 * \sum Y - \sum X * \sum XY}{n \sum X^2 - (\sum X)^2}$$

$$B = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

Example – What does that means??

Weight (lbs)	Height (inches)
140	60
155	62
159	67
179	70
192	71
200	72
212	75

Step 1: Calculate X*Y, X^2, and Y^2

Weight (lbs)	Height (inches)	X*Y	X^2	Y^2
140	60	8400	19600	3600
155	62	9610	24025	3844
159	67	10653	25281	4489
179	70	12530	32041	4900
192	71	13632	36864	5041
200	72	14400	40000	5184
212	75	15900	44944	5625

Step 2: Calculate ΣX, ΣY, ΣX*Y, ΣX^2, and ΣY^2

	Weight (lbs)	Height (inches)	X*Y	X^2	Y^2
	140	60	8400	19600	3600
	155	62	9610	24025	3844
	159	67	10653	25281	4489
	179	70	12530	32041	4900
	192	71	13632	36864	5041
	200	72	14400	40000	5184
	212	75	15900	44944	5625
Σ	**1237**	**477**	**85125**	**222755**	**32683**

Step 3: Calculate b_0 (our a)

- The formula to calculate a is: $[(\Sigma Y)(\Sigma X^2) - (\Sigma X)(\Sigma XY)] \ / \ [n(\Sigma X^2) - (\Sigma X)^2]$
- In this example, $a = [(477)(222755) - (1237)(85125)] \ / \ [7(222755) - (1237)^2] =$ **32.783**

Step 4: Calculate b_1

- The formula to calculate b_1 is: $[n(\Sigma XY) - (\Sigma X)(\Sigma Y)] \ / \ [n(\Sigma X^2) - (\Sigma X)^2]$
- In this example, $b_1 = [7(85125) - (1237)(477)] \ / \ [7(222755) - (1237)^2] =$ **0.2001**

Step 5: Place b_0 and b_1 in the estimated linear regression equation.

- The estimated linear regression equation is: $\hat{y} = b_0 + b_1 * x$
- In our example, it is $\hat{\mathbf{y}} = \mathbf{0.32783 + (0.2001) * x}$

How to Interpret a Simple Linear Regression Equation

- Here is how to interpret this estimated linear regression equation: $\hat{y} = 32.783 + 0.2001x$
- **b_0 = 32.7830**. When weight is zero pounds, the predicted height is 32.783 inches. Sometimes the value for b_0 can be useful to know, but in this example it doesn't actually make sense to interpret b_0 since a person can't weigh zero pounds.
- **b_1 = 0.2001**. A one pound increase in weight is associated with a 0.2001 inch increase in height.